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Method for manufacturing a patient-specific prosthesis for a fractured long bone

Abstract

Provided is a method of manufacturing a prosthesis for a fractured long bone of a patient, the method including the steps of: A) providing data representative of the fractured long bone in a patient; B) based on said data, designing the prosthesis specifically to the patient, the prosthesis including a stem part that is configured to secure fragment(s) of the fractured long bone at chosen securing position(s) that will apply chosen mechanical stress onto the bone fragments and reduce the risk of osteonecrosis of the bone fragments.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

(1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 C.F.R. § 1.57.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The invention relates to a method for manufacturing a patient-specific prosthesis for a fractured long bone of a patient and to a patient-specific prosthesis for a fractured long bone of a particular patient.

Description of the Related Art

(3) US 2004/0230311 A1 discloses a shoulder prosthesis comprising a stem to be inserted into the canal of the diaphyseal fragment of the humerus of a patient, an intermediary part reduced to a medial pillar and a head which is a generally spherical hollow cap. This type of shoulder prosthesis allows reattaching tuberosity fragments and humeral head fragments of the original humerus to the prosthesis. For reducing the risk of osteonecrosis of the reattached bone fragments and diaphyseal fragment, it is preferable that these fragments are mechanically loaded in the patient's body. The known shoulder prosthesis do not always ensure that every fragment is appropriately mechanically loaded, since the shape and the size of the fragments differ from one patient to another and do not always correspond to the bone fragments.

SUMMARY OF THE INVENTION

(4) The invention aims to solve the deficiencies of the abovementioned prior art.

(5) An aim of the invention is to provide a new method for manufacturing a prosthesis for a fractured long bone of a patient, implying fewer chances of osteonecrosis of the reattached bone fragments after implantation into the patient's body.

(6) The invention is defined below:

(7) A method for manufacturing a prosthesis for a fractured long bone of a patient, the method comprising the steps of: A) providing data representative of the fractured long bone of the patient, the fractured long bone comprising a diaphyseal fragment comprising a medullary cavity; B) based on the data provided at step A), designing the prosthesis specifically to the patient, the prosthesis comprising a stem part configured to be inserted into the medullary cavity for securing the stem part to the diaphyseal fragment, step B) comprising: a sub-step of choosing, specifically to the patient, at least one contact zone of the medullary cavity onto which a respective chosen mechanical stress is planned to be applied by the stem part, when the stem part is inserted into the medullary cavity and a sub-step of designing the stem part so that the stem part may be inserted into the medullary cavity and thus apply the chosen mechanical stress to said at least one contact zone; and C) manufacturing the prosthesis including the stem part designed at step B).

(8) In the invention, the data representative of the fractured long bone of the patient (step A) is used for designing (step B) and manufacturing (step C) a patient-specific prosthesis including a stem part with patient-specific features that allow, specifically to the patient, distributing mechanical loads onto the bone fragment, when the prosthesis is inserted into the patient's body. In other words, the prosthesis is manufactured with the most appropriate shape, tailored to the bone fragment. Thus, a patient-specific mechanical stress is applied to the bone fragment when the prosthesis is introduced into the patient's body and during healing of the fracture. The mechanical stress effectively applied to the bone fragment corresponds to conditions that may be planned in advance, prior to the step C of manufacturing the prosthesis. In particular, before the step of manufacturing, the surgeon may choose or plan, specifically to a particular patient, how the bone fragment will be submitted to mechanical stress when the prosthesis is implemented in the patient's body, and the prosthesis is manufactured in a way that this mechanical stress is actually achieved. The risk of post-surgery osteonecrosis is highly reduced, since the bone fragment is mechanically loaded exactly according to the needs of the patient's body.

(9) The stem part may have the most appropriate shape, tailored to be nested into the medullary cavity of the diaphyseal fragment of the patient to whom the prosthesis is intended, although the shape, the bone density and/or other parameters of the diaphyseal fragment may differ from one patient to another. Thanks to its patient-specific nesting shape, the stem part is sure to apply appropriate mechanical stress onto one or more contact zones of the medullary cavity. The mechanical stress to be applied can be chosen in advance, for example by the surgeon, as the patient's data is provided at step A), prior to designing the prosthesis (step B). Thus, the risk of osteonecrosis is reduced after the prosthesis is implemented into the patient.

(10) Further optional and advantageous features of the invention are defined below: said at least

one contact zone of the medullary cavity continuously covers a circumference of the medullary cavity. during sub-step B1), the chosen mechanical stress is chosen to be evenly distributed onto each of said at least one contact zone of the medullary cavity. step B) comprises the further sub-step of, based on the data provided at step A), designing the stem part so that the stem part comprises, specifically to the patient: a coverable diaphyseal portion, configured to be covered by the diaphyseal fragment when the stem part is secured to the diaphyseal fragment, whereas the rest of the stem part is left uncovered by the diaphyseal fragment; and a first visible mark, visually delimiting the coverable diaphyseal portion from the rest of the stem part. step A) comprises providing data representative of epiphyseal fragments of the fractured long bone; and step B) comprises, based on the data provided at step A), designing the prosthesis so that the stem part is configured for securing at least one of the epiphyseal fragments to the stem part. step B) comprises: a sub-step of choosing, specifically to the patient, respective securing positions of the epiphyseal fragments relative to each other, representative of how the epiphyseal fragments are planned to be positioned relative to each other when secured to the stem part, the securing positions being chosen so that a respective chosen mechanical stress is applied onto each epiphyseal fragment by at least one of the other epiphyseal fragments, when the epiphyseal fragments are secured at the respective chosen securing positions of the stem part, and a sub-step of designing the stem part so that the stem part is configured for securing the epiphyseal fragments at the respective chosen securing positions. step B) comprises the further sub-step of designing the stem part so that the stem part comprises a second visible mark indicating the respective securing positions of the epiphyseal fragments on the stem part. step B) comprises the further sub-step of designing the stem part based on the data provided at step A), so that the stem part comprises, specifically to the patient, coverable epiphyseal portions, being configured for being covered respectively by the epiphyseal fragments when the epiphyseal fragments are secured to the stem part at the respective securing positions; wherein the second visible mark is designed so as to visually delimit the coverable epiphyseal portions from each other. the stem part comprises, for at least one of the epiphyseal fragments, a respective plug, for securing the concerned epiphyseal fragment to the stem part, said at least one plug applying mechanical stress onto the epiphyseal fragment secured thereto. the epiphyseal fragments comprise: at least one viable tuberosity fragment, secured to a muscle of the patient, the muscle being attached to said at least one viable tuberosity fragment by means of a tendon of the muscle and a damaged articular fragment, initially being part of an damaged joint of the patient, for articulating the long bone with an auxiliary bone of the patient; step B) comprises: based on the data provided at step A), designing the prosthesis so that: the stem part is configured for securing said at least one viable tuberosity fragment to the stem part, and the prosthesis further comprises a head part, being configured to be secured to the stem part in replacement for the damaged articular fragment of the fractured long bone of the patient, a sub-step of choosing, specifically to the patient, respective securing positions of said at least one viable tuberosity fragment and of the head part relative to each other, representative of how said at least one viable tuberosity fragment and head part are planned to be positioned relative to each other when secured to the stem part, the securing positions being chosen so that a respective chosen mechanical stress is applied to each of said at least one viable tuberosity fragment by the head part, when said at least one viable tuberosity fragment and the head part are secured at the respective chosen securing positions of the stem part, and a sub-step of designing the stem part so that said at least one viable tuberosity fragment and the head part may be secured to the stem part at the respective chosen securing positions; step C) comprises providing or manufacturing the head part of the prosthesis. the head part comprises: a standard cap, comprising an articular surface of concave or convex shape for forming a prosthetic joint for replacement of the damaged joint of the patient, and comprising a securing surface opposed to the articular surface, the standard cap being secured to the stem part by means of the securing surface; a patient-specific insert, designed during step B) patient specifically, and manufactured during step C), configured to: be interposed between the

standard cap and the stem part, and apply the respective mechanical stress onto each of said at least one viable tuberosity fragment. step A) includes using CT scans of the patient and step C) includes additive manufacturing of at least a part of the prosthesis. the fractured long bone is a fractured humerus, the prosthesis being a shoulder prosthesis.

(11) Another object of the invention is defined below:

(12) A patient-specific prosthesis for a fractured long bone of a particular patient, the prosthesis comprising a stem part that is designed based on data representative of the fractured long bone of this particular patient, so as to: be inserted into the medullary cavity of a diaphyseal fragment of the fractured long bone of the patient for securing the stem part to the diaphyseal fragment, and apply a chosen mechanical stress to at least one contact zone of the medullary cavity, chosen based on said data.

(13) Further optional and advantageous features of the invention are defined below: the stem part comprises, specifically to the patient: a coverable diaphyseal portion, including at least a part of the stem part and being configured to be covered by the diaphyseal fragment when the stem part is inserted into the medullary cavity, whereas the rest of the stem part is left uncovered by the diaphyseal fragment; and a first visible mark, visually delimiting the coverable diaphyseal portion from the rest of the stem part.

(14) An aim of the invention is to provide a new method for manufacturing a prosthesis for a fractured long bone of a patient, implying fewer chances of osteonecrosis of the reattached bone fragments after implantation into the patient's body.

(15) The invention is defined as follows:

(16) A method for manufacturing a prosthesis for a fractured long bone of a patient, the method comprising the steps of: A) providing data representative of the fractured long bone of the patient, the fractured long bone comprising epiphyseal fragments, each epiphyseal fragment preferably being: either a tuberosity fragment, secured to a muscle of the patient, the muscle being attached to the tuberosity fragment by means of a tendon of the muscle, or an articular fragment, being part of a joint of the patient, for articulating the long bone with an auxiliary bone of the patient; B) based on the data provided at step A), designing the prosthesis specifically to the patient, the prosthesis comprising a stem part configured for securing the epiphyseal fragments to the stem part, step B) comprising: a sub-step of choosing, specifically to the patient, respective securing positions of the epiphyseal fragments relative to each other, representative of how the epiphyseal fragments are planned to be positioned relative to each other when secured to the stem part, the securing positions being chosen so that a respective chosen mechanical stress is applied onto each epiphyseal fragment by at least one of the other epiphyseal fragments, when the epiphyseal fragments are secured at the respective chosen securing positions of the stem part, and a sub-step of designing the stem part so that the stem part is configured for securing the epiphyseal fragments at the respective chosen securing positions; and C) manufacturing the prosthesis including the stem part designed at step B).

(17) In the invention, the data representative of the fractured long bone of the patient (step A) is used for designing (step B) and manufacturing (step C) a patient-specific prosthesis including a stem part with patient-specific features that allow, specifically to the patient, distributing mechanical loads onto the bone fragments, when the prosthesis is inserted into the patient's body. In other words, the prosthesis is manufactured with the most appropriate shape, tailored to the bone fragments. Thus, a patient-specific mechanical stress is applied to the bone fragments when the prosthesis is introduced into the patient's body and during healing of the fracture. The mechanical stress effectively applied to the bone fragments corresponds to conditions that may be planned in advance, prior to the step C of manufacturing the prosthesis. In particular, before the step of manufacturing, the surgeon may choose or plan, specifically to a particular patient, how the bone fragments will be submitted to mechanical stress when the prosthesis is implemented in the patient's body, and the prosthesis is manufactured in a way that this mechanical stress is actually

achieved. The risk of post-surgery osteonecrosis is highly reduced, since the bone fragments are mechanically loaded exactly according to the needs of the patient's body.

(18) The stem part is built so that the epiphyseal fragments may be reattached at predetermined respective securing positions by the surgeon. Preferably, the stem part is configured with a patient-specific shape and features, for obtaining that the epiphyseal fragments can only be reattached at these respective securing positions. When positioned with this scheme, a patient-specific mechanical stress is applied by the bone fragments to each other. In this method, the stress applied on the bone fragments is chosen prior to designing the prosthesis, thanks to the data representative of the patient. This method ensures that every manufactured prosthesis, intended for a respective patient, will apply the intended mechanical stress to the fragments, even if the fragments differ from one patient to the other, in terms of shape, bone density or other parameters. This allows reducing the risk of osteonecrosis when the prosthesis is implemented in the patient.

(19) Further optional and advantageous features of the invention are defined below: Step B) comprises the further sub-step of designing the stem part so that the stem part comprises a visible mark indicating the respective securing positions of the epiphyseal fragments on the stem part. Step B) comprises the further sub-step of designing the stem part based on the data provided at step A), so that the stem part comprises, specifically to the patient, coverable epiphyseal portions, being configured for being covered respectively by the epiphyseal fragments when the epiphyseal fragments are secured to the stem part at the respective chosen positions; wherein the visible mark is designed so as to visually delimit the coverable epiphyseal portions from each other. The stem part comprises, for at least one of the epiphyseal fragments, a respective plug, for securing the concerned epiphyseal fragment to the stem part, said at least one plug applying mechanical stress onto the epiphyseal fragment secured thereto.

(20) Another object of the invention is defined as follows:

(21) A patient-specific prosthesis for a fractured long bone of a particular patient, the fractured long bone comprising epiphyseal fragments, the prosthesis comprising a stem part that is designed based on data representative of the fractured long bone of this particular patient, the stem part being configured so that each epiphyseal fragment may be secured to the stem part at a chosen securing position relative to the other epiphyseal fragments, the securing positions being chosen based on said data so that, when the epiphyseal fragments are secured at the respective securing positions, a respective chosen mechanical stress is applied onto each epiphyseal fragment by at least one of the other epiphyseal fragments.

(22) Further optional and advantageous features of the invention are defined below: The stem part comprises, specifically to the patient: a visible mark indicating the respective securing positions of the epiphyseal fragments on the stem part; and coverable epiphyseal portions, being configured for being covered respectively by the epiphyseal fragments when the epiphyseal fragments are secured to the stem part at the respective securing positions; wherein the visible mark is designed so as to visually delimit the coverable epiphyseal portions from each other. The stem part comprises, for at least one of the epiphyseal fragments, a respective plug, for securing the concerned epiphyseal fragment to the stem part, said at least one plug applying mechanical stress onto the epiphyseal fragment secured thereto.

(23) An aim of the invention is to provide a new method for manufacturing a prosthesis for a fractured long bone of a patient, implying fewer chances of osteonecrosis of the reattached bone fragments after implantation into the patient's body.

(24) The invention is defined as follows:

(25) A method for manufacturing a prosthesis for a fractured long bone of a patient, the method comprising the steps of: A) providing data representative of the fractured long bone of the patient, the fractured long bone comprising: at least one viable tuberosity fragment, secured to a muscle of the patient, the muscle being attached to said at least one viable tuberosity fragment by means of a tendon of the muscle and a damaged articular fragment, initially being part of a damaged joint of

the patient, for articulating the long bone with an auxiliary bone of the patient; B) based on the data provided at step A), designing the prosthesis specifically to the patient, the prosthesis comprising: a stem part, configured for securing said at least one viable tuberosity fragment to the stem part, and a head part, being configured to be secured to the stem part in replacement for the damaged articular fragment, step B) comprising: a sub-step of choosing, specifically to the patient, respective securing positions of said at least one viable tuberosity fragment and of the head part relative to each other, representative of how said at least one viable tuberosity fragment and head part are planned to be positioned relative to each other when secured to the stem part, the securing positions being chosen so that a respective chosen mechanical stress is applied to each of said at least one viable tuberosity fragment by the head part, when said at least one viable tuberosity fragment and the head part are secured at the respective chosen securing positions of the stem part, and a sub-step of designing the stem part so that said at least one viable tuberosity fragment and the head part may be secured to the stem part at the respective chosen securing positions; and C) manufacturing the prosthesis including the stem part designed at step B) and providing or manufacturing the head part of the prosthesis.

(26) In the invention, the data representative of the fractured long bone of the patient (step A) is used for designing (step B) and manufacturing (step C) a patient-specific prosthesis including a stem part and a head part with patient-specific features that allow, specifically to the patient, distributing mechanical loads onto the bone fragments, when the prosthesis is inserted into the patient's body. In other words, the prosthesis is manufactured with the most appropriate shape, tailored to the bone fragments. Thus, a patient-specific mechanical stress is applied to the bone fragments when the prosthesis is introduced into the patient's body and during healing of the fracture. The mechanical stress effectively applied to the bone fragments corresponds to conditions that may be planned in advance, prior to the step C of manufacturing the prosthesis. In particular, before the step of manufacturing, the surgeon may choose or plan, specifically to a particular patient, how the bone fragments will be submitted to mechanical stress when the prosthesis is implemented in the patient's body, and the prosthesis is manufactured in a way that this mechanical stress is actually achieved. The risk of post-surgery osteonecrosis is highly reduced, since the bone fragments are mechanically loaded exactly according to the needs of the patient's body.

(27) In the invention, a head part of standard design, or at least partially of patient-specific design, is used for replacing the damaged fragment and for applying a chosen mechanical stress onto the viable tuberosity fragments. Also, the stem part, including the securing positions, is designed so that the viable fragments and the head part may be positioned, or must be positioned, so that the chosen mechanical stress is actually applied when the prosthesis is implanted within the patient. The head part and the stem part, including the securing positions, are designed specifically to one particular patient, based on the data provided at step A), so that mechanical stress tailored to the patient is applied when the prosthesis manufactured at step C) is actually positioned within the patient. This method ensures that every manufactured prosthesis, intended for a respective patient, will apply the intended mechanical stress to the fragments, even if the fragments differ from one patient to the other, in terms of shape, bone density or other parameters. This allows reducing the risk of osteonecrosis when the prosthesis is implemented in the patient.

(28) Further optional and advantageous features of the invention are defined below:

(29) The head part comprises: a standard cap, comprising an articular surface of concave or convex shape for forming a prosthetic joint for replacement of the damaged joint of the patient, and comprising a securing surface opposed to the articular surface, the standard cap being secured to the stem part by means of the securing surface; and a patient-specific insert, designed during step B) patient specifically, and manufactured during step C), configured to: be interposed between the standard cap and the stem part, and apply the respective mechanical stress onto each of said at least one viable tuberosity fragment.

(30) Another object of the invention is defined as follows:

(31) A patient-specific prosthesis for a fractured long bone of a particular patient, the fractured long bone comprising: at least one viable tuberosity fragment, secured to a muscle of the patient by means of a tendon of the muscle, a damaged articular fragment, initially being part of a damaged joint of the patient, for articulating the long bone with an auxiliary bone of the patient; wherein the prosthesis comprises: a stem part, configured for securing said at least one viable tuberosity fragment to the stem part, and a head part, being configured to be secured to the stem part in replacement for the damaged articular fragment, wherein the stem part is designed based on data representative of the fractured long bone of this particular patient, the stem part comprising securing positions, chosen specifically to this patient based on said data, so that the viable tuberosity fragment and the head part may each be secured to the stem part at a respective securing position, the securing positions being chosen so that, when said at least one viable tuberosity fragment and the head part are secured at the securing positions, a respective chosen mechanical stress is applied onto each viable tuberosity fragment by the head part.

(32) Further optional and advantageous features of the invention are defined below:

(33) The head part comprises: a standard cap, comprising an articular surface of concave or convex shape for forming a prosthetic joint for replacement of the damaged joint of the patient, and comprising a securing surface opposed to the articular surface, the standard cap being secured to the stem part by means of the securing surface; a patient-specific insert, designed patient specifically and configured to: be interposed between the standard cap and the stem part, and apply the respective mechanical stress onto each of said at least one viable tuberosity fragment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further advantages and advantageous features of the invention are disclosed in the following description, provided in reference to the appended drawings, solely for exemplary non-limitative purpose.

(2) In the drawings, FIG. 1 is an exploded view of several embodiments.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(3) A fractured long bone of the patient is depicted on FIG. 1. In the present case, the fractured long bone is a humerus, broken in four fragments at its shoulder end. However, the invention also applies to other long bones of the body, such as the hip end of a femoral bone.

(4) The invention is preferably applied to a fractured long bone of a human patient. However, it may be applied to a fractured long bone of an animal patient.

(5) The depicted fractured long bone comprises a diaphyseal fragment 2, or shaft fragment. The diaphyseal fragment 2 includes essentially the shaft part of the original bone.

(6) The fragment 2 defines a diaphyseal axis X2, which is extending along the fragment 2. This fragment 2 is of generally tubular shape around axis X2. The fragment 2 comprises a medullary cavity 8, extending along axis X2, in particular coaxially. The cavity 8 is opened at a proximal end 10 of the fragment 2, where the bone is fractured. In the present case of a humeral diaphyseal fragment 2, the proximal end 10 is defined along axis X2 towards the shoulder joint of the patient, opposite to a distal end of the fragment 2 directed towards the elbow joint of the patient (not shown).

(7) At the proximal end 10, the fragment 2 forms a fracture line 9, delineating the open cavity 8. The fracture line 9 surrounds the axis X2.

(8) The depicted fractured long bone further comprises three epiphyseal fragments 4, 5 and 6. At least some of these fragments 4, 5 and 6 are to be reattached at the end 10 of the fragment 2 for reconstructing the original bone.

(9) Each fragment 4, 5 and 6 respectively has an internal surface 27, 28 and 29, as well as an

external surface **23**, **24** and **25**. The surface **27** of the fragment **4** is opposed to the surface **23**. The surface **27** is delimited from the surface **23** by a fracture line **41** of the fragment **4**, surrounding surface **23** and surface **27**. The surface **28** of the fragment **5** is opposed to the surface **24**. The surface **28** is delimited from the surface **24** by a fracture line **42** of the fragment **5**, surrounding surface **23** and surface **27**. The surface **29** of the fragment **6** is opposed to the surface **25**. The surface **29** is delimited from the surface **25** by a fracture line **43** of the fragment **6**, surrounding surface **25** and surface **29**.

(10) Depending on the fracture type and on the long bone considered, more or less than three epiphyseal fragments may be formed from the initial end of the patient's long bone. Only one epiphyseal fragment may be formed. However, the present invention preferably applies to cases where more than one epiphyseal fragments are formed.

(11) Some of the epiphyseal fragments, like fragments **4** and **5** of the present example, may be designated as “tuberosity fragments”. These particular fragments are each secured to a muscle of the patient by means of a tendon of the concerned muscle. In other words, the implantation sites of the concerned muscles are located on these tuberosity fragments.

(12) A muscle **7** is implanted to the external surface **23** of the fragment **4**. In the present case, the fragment **4** includes the greater tuberosity of the original bone, originally fixed with a partially-illustrated supraspinatus muscle **7** of the patient.

(13) A muscle **13** is implanted to the external surface **24** of the fragment **5**. In the present case, the fragment **5** includes the lesser tuberosity of the original bone, originally fixed with a partially-illustrated subscapularis muscle **13** of the patient.

(14) Some other epiphyseal fragments, like fragment **6** of the present example, may be designated as “articular fragment”. Each articular fragment is initially part of a joint of the patient, for articulating the long bone with an auxiliary bone of the patient.

(15) In the present case, the articular fragment **6** is a humeral head of the humerus, which is originally part of the shoulder joint, for articulating the humerus with a scapula of the patient (not shown on the drawings). In this case, the scapula forms the “auxiliary bone” of the shoulder joint. More precisely, the external surface **25** of the humeral head **6** has a generally spherical and convex shape and is initially articulated with a corresponding concave surface of a glenoid part of the scapula.

(16) Should some fragments have cracks or have geometrical defects may be spatially reconstructed with bone graft, cement or any other suitable replacement material. In the present example, fragment **4** comprises a portion **70** of replacement material for filling a notch of the fragment **4**. In this case, the portion **70** forms a part of the edge of the fragment **4**, considered as a part of the fracture line **41**, for the sake of simplicity.

(17) As visible on FIG. **1**, a prosthesis **11** comprises a stem part **12** and advantageously a head part **14**. In the case the fractured long bone is a humerus, the prosthesis **11** constitutes a shoulder prosthesis, or at least a humeral component of a shoulder prosthesis. This prosthesis **11**, at least the stem part **12**, is patient-specific. In other words, the prosthesis **11** is designed depending on the geometry of the fractured long bone of one specific patient, the prosthesis **11** being intended to be implanted into this particular patient.

(18) The stem part **12** comprises: a coverable diaphyseal portion **16**, forming a distal end of the stem part **12** along axis **X12**; and coverable epiphyseal portions **20**, **21** and **22**, located at a proximal end of the stem part **12** along axis **X12**.

(19) In some embodiments, the portions **16**, **20**, **21** and **22** form a single integral piece. In some other embodiments, one or more portions may be a separate piece assembled with the others.

(20) The stem part **12** is configured to be inserted into the medullary cavity **8** of the diaphyseal fragment **2**, through the open proximal end **10**, as depicted with the arrow **A8**, parallel to axis **X2**. The stem part **12** preferably defines an axis **X12**, that is parallel or coaxial with axis **X2** when the stem part **12** is secured to the fragment **2**. This insertion of the stem part **12** preferably ensures

securing the stem part **12** to the fragment **2**. In some embodiments, supplementary means may be used for reinforcing the securing, such as cement, fasteners, or other conventional means.

(21) The coverable diaphyseal portion **16** is configured for securing the stem part **12** to the fragment **2**, preferably by fitting or anchoring of the portion **16** into the medullary cavity **8**. At least, the portion **16** has a shape that corresponds to the shape of the medullary cavity **8**, specifically to this patient. The portion **16** is configured so that its exterior surface is covered entirely by the diaphyseal fragment **2** when the stem part **12** is secured to the diaphyseal fragment **2**, whereas the rest of the stem part **12**, located at a proximal end of the stem part **12**, is left uncovered by the diaphyseal fragment **2**. In this case, “the rest of the stem part **12**” includes the coverable epiphyseal portions **20**, **22** and **24**.

(22) When inserted into the medullary cavity **8**, the stem part **12** applies mechanical stress to the fragment **2**, to an extent than may be chosen in advance prior or during designing of the prosthesis **11**, thanks to a method explained below. In the present example, the stem part **12** is configured so that the predetermined mechanical stress is applied to a continuous contact zone **40** of the fragment **2**, also chosen in advance. This zone **40** is formed at the surface thereof, inside the cavity **8**, as depicted on FIG. **1**. This zone **40** is entirely in contact with the portion **16** of the stem part **12**.

(23) The zone **40** preferably extends all around axis X2, drawing a ring-like shape. In other words, the zone extends continuously along a circumference of the medullary cavity around axis X2.

(24) In other embodiment, the mechanical stress may be applied to several distinct zones of the cavity **8**.

(25) In any case, the amount of mechanical stress to be applied is chosen in advance, the contact zone is chosen in advance, and the prosthesis is manufactured so that said chosen mechanical stress is actually applied onto said chosen contact zone **40** when the prosthesis **11** is secured to the fragment **2**. Thus, the prosthesis is made patient-specifically. An appropriate mechanical stress may be applied onto the diaphyseal fragment **2** so as to avoid osteonecrosis, or any other consequence relative to a lack of osseous mechanical loading.

(26) In the present invention, by “mechanical stress” applied to a bone fragment, it is meant a mechanical stress sufficient to avoid osseous necrosis. The mechanical stress applied to the bone fragments is also chosen not to exceed the mechanical resistance of the bone fragments. In other words, the mechanical stress is chosen not to break or fracture the bone fragments. The appropriate stress to be applied may be calculated depending on the geometry of the considered bone fragments and on the osseous density of these fragments. For example, the mechanical stress may be a pressure exerted onto the considered bone fragment, traction, flexion or the like.

(27) Preferably, the mechanical stress is distributed evenly on the contact zone **40**, or on the contact zones if several contact zones are foreseen. In other words, a same amount of stress is applied to any part of the contact zone **40** or zones. Preferably, the mechanical stress is applied in radial outward directions around axis X12 by the coverable portion **16** onto the zone **40**.

(28) In order to be able to apply the appropriate amount of stress onto the contact zone and to choose what zone of the medullar cavity should serve as the contact zone, a specific method of manufacturing is performed for obtaining the patient-specific prosthesis **11**.

(29) For each prosthesis to be obtained and each patient to be treated, the method comprises a preliminary step A of providing data representative of the fractured long bone of this patient, including essentially information relative to geometry of the diaphyseal fragment **2**. The information may also be relative to the osseous density of the fragment **2**.

(30) Preferably, this step of providing data is at least partially achieved by scanning the relevant part of the patient including the fractured long bone, for example with a method of CT-scan (“computerized tomography”).

(31) Successively to step A, based on the data gathered at step A, a step B of the method is performed. This step B includes a sub-step of choosing: which zone or zones of the medullary cavity **8** need to be stressed, and the magnitude of the stress to be applied onto said zone.

(32) Once the zone and the magnitude of the stress are chosen, the stem part **12** of the prosthesis is designed accordingly, during a subsequent sub-step of step B. The designed stem part **12** is designed so as to be in contact with the chosen zone, becoming the aforementioned contact zone **40**, so as to apply the chosen pressure, when effectively mounted to the epiphyseal fragment.

(33) Thus, this step B allows designing the prosthesis **11**, including in particular the stem part **12**, specifically to the patient intended to receive this prosthesis **11**.

(34) During a step C of the method, successive to the step B, the prosthesis **11** is manufactured, including the stem part **12** as it was designed during the second step. Thus, the patient-specific prosthesis **11** is obtained, for a specific patient.

(35) The step C of manufacturing preferably includes additive manufacturing of the entire stem part **12**. In some embodiment, only portions of the stem part **12** that are required to be patient-specific, such as the exterior surface in contact with the contact zone **40**, are manufactured by additive manufacturing, these patient specific portion being combined with standard portions for forming the stem part **12**.

(36) For the present invention, any part obtained by additive manufacturing may be metallic. Instead, appropriate plastic material may be used.

(37) Preferably, the stem part **12** comprises a visible mark **18** as visible on FIG. **1**. The visible mark **18** is formed at the exterior surface of the stem part **12**. The mark **18** may be a shallow carving, an embossed marking, a colored marking, or the combination thereof, on the surface of the stem part **12**. In any case, the mark **18** is configured to be visible to the eye of the surgeon during surgery. The mark **18** preferably forms a line, continuous or dashed. Instead, the mark **18** may form a dot, or several dots.

(38) The mark **18** visually delimits the coverable diaphyseal portion **16** from the rest of the stem part **12**. The prosthesis **11**, in particular the coverable portion **16** and the mark **18**, is designed specifically to the patient so that, when the stem part **12** is inserted into the cavity **8** properly, in particular at the right position relative to the fragment **2** along the diaphyseal axis X2, the mark **18** and the fracture line **9** are superposed. Thus, during surgery, the surgeon is informed by the mark **18** whether the stem part **12** is properly positioned into the cavity **8**, at least concerning the position of the stem part **12** relative to the fragment **2** along the axis X2. Also, the shape of the mark **18** may indicate to the surgeon whether the stem part **12** is properly positioned relative to fragment **2**, around axis X2. For example, the mark **18** may indicate with a dot, or have a portion reproducing the shape of, a differential pattern around axis X2, such as a dent or a notch, of the fracture line **9**. The surgeon has to ensure that the dot or the portion is aligned with the dent.

(39) Preferably, a correct position of the stem part **12** is obtained when the mark **18** and the fracture line **9** are aligned and/or superposed. In case of misalignment, the surgeon may adjust the position of the stem part **12** relative to the fragment **2** during surgery. Thus, the mark **18** promotes adequate positioning of the stem part **12**, and makes surgery easier. In this adequate positioning, the chosen stress is sure to be applied to the chosen contact zone **40** by the stem part **12** onto the fragment **2**.

(40) For obtaining the mark **18** disclosed above, the data of the patient provided during the preliminary step A) of the method is used. During the step B) of designing the prosthesis **11**, the stem part **12** is designed including the coverable diaphyseal portion **16** and the mark **18**, so that they may achieve the above-disclosed functions. In particular, the stem part **12** is designed so that the mark **18** delimits the coverable diaphyseal portion **16** and/or indicates where the fracture line **9** is foreseen to be located when the stem part **12** is inserted into the medullary cavity **8**, covers the portion **16** and the stem part **12** applies the chosen stress to the zone **40**.

(41) In an embodiment, the portion **16** of the stem part **12** is radially outwardly expansible around axis X12. The expansion may be activated by the surgeon. For example, the stem part **12** is designed and manufactured including an expansion screw **60**, coaxial with axis X12. The screw may be actuated from outside of the portion **16** by the surgeon: for example the screw head is accessible from the opposite end of the stem part **12**. The stem part **12** may be introduced into the

cavity **8** to the desired position, and then, the surgeon may actuate the screw **60** for expanding the stem part radially outwardly, so that the stem part applies the chosen mechanical stress onto the zone **40**. Any other suitable stem part expansion actuator than the screw **60** may be provided instead.

(42) The stem part **12** is configured for receiving the epiphyseal fragments **4**, **5** and **6**. In other words, these fragments **4**, **5** and **6** may be secured to the stem part **12**.

(43) Specifically to the patient, the stem part **12** is designed so as to ensure a patient-specific positioning of the fragments **4**, **5** and **6** relative to the stem part **12** and to each other. The positioning of the fragment **2** relative to the stem part **12** may also be planned patient-specifically, as explained above, so that eventually, the fragments **2**, **4**, **5** and **6** are positioned relative to each other in a chosen manner when secured to the stem part **12**. As concerns the fragments **4**, **5** and **6**, the stem part **12** is designed so that the fragments are positioned at chosen securing positions, defined in advance, specifically to the patient. In particular, this positioning is chosen specifically to the shape of the fragments **4**, **5** and **6**, and preferably also to the shape of the fragment **2**.

(44) In the method of manufacturing the prosthesis **11**, step A) may include providing data relative to the fragments **4**, **5** and **6** of the long bone of the patient. This may be performed alternatively or additionally to providing data relative to the fragment **2**. The data provided may be relative to the shape of the fragments **4**, **5** and **6**. The data may also be relative to the osseous density of the fragments **4**, **5** and **6**. The data for fragments **4**, **5** and **6** may be obtained by CT scanning, as explained above for fragment **2**.

(45) Step B) may include a sub-step of choosing, specifically to the patient, respective securing positions of the fragments **4**, **5** and **6**. The chosen securing positions illustrate how the fragments **4**, **5** and **6** will be positioned relative to each other when secured to the stem part **12**. The chosen securing positions may depend from the actual size and shapes of the fragments **4**, **5** and **6** of the considered patient.

(46) This sub-step of choosing the securing positions may be performed alternatively or additionally to the sub-step of choosing the contact zone **40** for fragment **2**.

(47) In this case, once the securing positions are chosen, the sub-step of designing the stem part **12** is performed, so that the stem part **12** is configured for securing the epiphyseal fragments at the respective chosen securing positions. In other words, the designed stem part enables or even imposes that, when the fragments **4**, **5** and **6** are secured thereto, they are in the planned positions.

(48) Then, during step C), the prosthesis **11**, including in particular the stem part **12** with the features designed in step B) above, is manufactured.

(49) The fragments **4**, **5** and **6** are configured to be positioned onto the stem part **12** as illustrated in FIG. **1** with the arrows **A4**, **A5** and **A6**, respectively.

(50) When the fragments **4**, **5** and **6** are secured to the stem part **12**, each coverable epiphyseal portion **20**, **21** and **22** is configured for being essentially covered, preferably completely covered, by one of the epiphyseal fragments **4**, **5** and **6**, respectively. Preferably, for receiving the fragments **4**, **5** or **6**, each respective coverable portion **20**, **21** and **22** has an external surface, shaped in correspondence to an internal surface **27**, **28** or **29** of the respective concerned fragment **4**, **5** or **6**. When received properly, namely according to the chosen securing position, each fragment **4**, **5** and **6** preferably entirely covers the exterior surface of the concerned portion **20**, **21** and **22**. For this purpose, the portions **20**, **21** and **22** of the stem part **12** are designed patient specifically during step B).

(51) Preferably, the stem part **12** is designed so that the reattached fragments **4**, **5** and **6** are positioned in their original position relative to each other and to the diaphyseal fragment **2**, as at the time when the bone was not yet fractured.

(52) In an alternative embodiment, the stem part **12** may be configured so that one or more of the fragment **2**, **4**, **5** and **6** is planned to be in a different position than its original position relative to the other fragments.

(53) Preferably, the stem part **12** is designed so that the fragments **2**, **4**, **5** and **6** bear against each other by means of their respective fracture lines **9**, **41**, **42** or **43** when they are secured to the stem part **12**. Thus, the chosen securing positions are preferably positions where the fragments **4**, **5** and **6** bear against each other, and optionally at least one of said fragments **4**, **5** and **6** bear against fragment **2**. At least two of the fragments **2**, **4**, **5** and **6** bear against each other in this manner. For example, when secured to the stem part **12**, fragment **4** is in abutting contact with fragment **5**, the fracture line **41** being in abutting contact with the fracture line **42**.

(54) In an embodiment, the stem part **12** is designed with a chosen securing position for the fragments **4**, **5** and **6**, enabling that the fragments **4**, **5** and **6**, and optionally fragment **2**, apply mechanical stress to each other when the fragments **2**, **4**, **5** and **6** are actually positioned at these chosen securing positions.

(55) For example, the stem part **12** is designed so that positioning the fragment **4** at its securing position on the stem part **12** implies that this fragment **4** is compressed between fragments **2**, **5** and **6**, if these fragments **2**, **5** and **6** are also positioned at their respective securing positions. In this case, the fragment **2**, **5** and **6** apply mechanical stress to the fragment **4** by means of their respective fracture lines **9**, **42** and **43**, onto the fracture line **41** of the fragment **4**. In this case, resulting mechanical stress is also applied onto the fragments **2**, **5** and **6**. This mechanical stress is achieved by planning an adequate positioning of the fragments **2**, **4**, **5** and **6** relative to each other.

Specifically, the stem part **12** may be designed so that the fragments **2**, **4**, **5** and **6** must be tightly fitted against each other by the surgeon when secured to the stem part **12** at their respective chosen securing positions. The amount of mechanical stress to be applied to each fragment **2**, **4**, **5** and **6** by one or more other of these fragments may also be chosen by appropriate designing of the stem part **12** and choice of positioning of the fragments.

(56) Thus, choosing the securing positions during step B) is made so that the abovementioned mechanical stress, with a chosen magnitude, is applied on the fragments **4**, **5** and **6** when actually positioned this way. Thus, the obtained prosthesis **11** avoids osteonecrosis by submitting the reattached fragment to an appropriate mechanical stress.

(57) For ensuring that the planned positioning of the fragments **4**, **5** and **6** is achieved, the exterior surface of the stem part **12**, in particular for the portions **20**, **21** and **22**, preferably has a shape corresponding to the shape of fragments **4**, **5** and **6**, in particular corresponding to the surfaces **27**, **28** and **29** of said fragments. The surgeon is informed that one epiphyseal fragment is correctly positioned, according to the chosen securing position, if the epiphyseal fragment fits onto the stem part **12**. In case of incorrect positioning of the epiphyseal fragment, said fragment does not fit with the stem part **12**.

(58) Additionally or alternatively, for ensuring that the chosen securing position of the fragments **4**, **5** and **6** is achieved, in the example illustrated on FIG. **1**, the stem part **12** may comprise three plugs **30**, **31** and **32**, each constituting a distinct securing element of one of the fragments **4**, **5** and **6**, respectively. Each plug **30**, **31** and **32** protrudes from the exterior surface of one of the portions **20**, **21** or **22**, respectively. Each plug is preferably frustoconical, pyramidal, or shaped as a dome. Conversely, each fragment **4**, **5** and **6** may have a blind bore, provided on its internal surface **27**, **28** and **29** respectively. These blind bores are preferably drilled or carved in the fragments **4**, **5** and **6** by the surgeon with a specific tool. The shape of each bore corresponds to the shape of the plugs **30**, **31** and **32**, so that each plug may be inserted into the bore of one fragment **4**, **5** or **6** when the concerned fragment **4**, **5** or **6** is positioned on the stem part **12**. Thus, when the plug is inserted into the appropriate bore of a given fragment **4**, **5** or **6**, the surgeon is sure that this fragment is correctly positioned onto the stem part **12**.

(59) Preferably, the method of manufacturing includes designing and manufacturing one or more patient-specific jigs, depending on the chosen securing position and on the data representative of the patient. Each jig, for example embodied as a drill guide, may be used by the surgeon for drilling the blind bore on one of the epiphyseal fragments. Each jig is adapted to the shape of the bone

fragments to be reworked by the surgeon. Thus, the surgeon is sure to drill the blind bores at the appropriate position on the fragments **4**, **5** and **6**, thus enabling a positioning of the fragments **4**, **5** and **6** at the chosen securing positions.

(60) In a preferable embodiment, the plugs are configured so as to apply mechanical stress onto the fragments **4**, **5** and **6**. For example one of the epiphyseal fragments may be compressed between the plug, on which it is mounted, and another epiphyseal fragment. In this example, the plug radially applies mechanical stress to the blind bore of the epiphyseal fragment and said fragment receives mechanical stress onto the fracture line in an opposed direction. Thus, the risk of osseous necrosis is reduced.

(61) Furthermore, the plugs may be designed so as to apply mechanical stress to the blind hole of the fragment by tight fitting into the blind hole.

(62) More than one plug may be provided for each epiphyseal portion, and a plurality of plugs may be provided for positioning each epiphyseal fragment.

(63) In some embodiments, the plugs are formed integral with the concerned epiphyseal portion. Alternatively, one or more of the securing element may be a separate part assembled with the concerned epiphyseal portion.

(64) In some embodiments, instead of the aforementioned plugs, any other type of positioning element may be provided.

(65) Additionally or alternatively, the stem part **12** comprises a visible mark **45** indicating the respective chosen securing positions of the epiphyseal fragments **4**, **5** and **6** on the stem part. The visible mark **45** is formed at the exterior surface of the stem part **12**. The mark **45** may be a shallow carving, an embossed marking, a colored marking, or the combination thereof, on the surface of the stem part **12**. In any case, the mark **45** is configured to be visible to the eye of the surgeon during surgery. The mark **45** preferably forms one or more lines, continuous or dashed. Instead, the mark **45** may form one or more dots.

(66) In a preferred embodiment, as illustrated on FIG. **1**, the mark **45** visually delimits the coverable epiphyseal portions **20**, **21** and **22** from each other and optionally from the rest of the stem part **12**. The coverable portions **20**, **21** and **22** and the mark **45** are designed specifically to the patient so that, when the fragments **4**, **5** and **6** of this patient are positioned at the planned position on the stem part **12**, the fracture lines **41**, **42** and **43** are superposed with the mark **45**. In this aspect, the chosen securing positions are indicated by the mark **45**. During surgery, the surgeon is informed by the mark **45** whether the fragments **4**, **5** and **6** are properly positioned on the stem part **12**.

(67) Thus, the mark **45** promotes adequate positioning of the fragments **4**, **5** and **6**, and makes surgery easier.

(68) For obtaining the mark **45** disclosed above, the data of the patient provided during the step A) of the method is used. During the step B), the stem part **12** is designed including the coverable portions **20**, **21**, **22** and the mark **45**, so that they may achieve the above-disclosed functions. In particular, the stem part **12** is designed so that the mark **45** delimits the coverable epiphyseal portions **20**, **21** and **22** from each other and/or indicates where the fracture lines **41**, **42** and **43** are foreseen to be located when the fragments **4**, **5** and **6** are positioned according to the chosen securing positions.

(69) For securing the fragments **4**, **5** and **6** to the stem part **12**, further conventional means suitable to the situation may be used, such as fasteners, cement and/or sutures.

(70) The surgeon may decide that the initial articular fragment **6** is reattached to the stem part **12**, as disclosed above. In this situation, the head part **14** is optional. For this situation, the prosthesis **11** may be provided without such head part **14**.

(71) In some specific cases, the surgeon may decide that some of the epiphyseal fragments are viable and may be reattached to the stem part **12** and some other fragments are damaged and may not be reattached. In this case, the damaged epiphyseal fragments may require to be replaced with prosthetic means.

(72) In the case the articular fragment **6** is damaged, while the fragments **4** and **5** are viable, the fragment **6** may be replaced by the prosthetic head part **14** shown on FIG. **1**. Thus, the head part **14** is configured to be secured to the prosthetic stem part **12** so as to replace the damaged articular fragment **6** when the prosthesis **11** is introduced in the patient's body. The head part **14** is preferably secured at portion **22** of the stem part **12**. The securing may be achieved with fasteners or any other suitable securing means. In the case the head part **14** is provided, the plug **32** is optional. Appropriate securing means may be provided on the portion **22** additionally or alternatively to plug **32**.

(73) In this case, the method comprises, during step A), providing data relative to the viable fragments **4** and **5** and to the damaged fragment **6**. Then, during step B), the prosthesis **11** is designed specifically to the patient, including the stem part **12** and the head part **14** being patient-specific.

(74) In the case a head part **14** is used instead of reattaching the fragment **6**, the method is similar than the previously explained method. Step B) includes a sub-step of choosing, specifically to the patient, respective securing positions of the viable fragments **4** and **5** and of the head part **14** relative to each other, and advantageously, relative to fragment **2**. In other words, in this sub-step, the position of the head part **14** is chosen, instead of the position of the reattached fragment **6** for the case explained above. The chosen "securing positions" are representative of how the viable fragments **4** and **5** and the head part **14** are planned to be effectively positioned relative to each other when secured to the stem part **12** by the surgeon. The fragments **4** and **5** and the head part **14** are configured to be positioned onto the stem part **12** as illustrated in FIG. **1** with the arrows A**4**, A**5** and A**14**, respectively. The fragments **4** and **5** are positioned as disclosed above, although fragment **6** is replaced by the head part **14**. Thus, the head part is positioned in a similar way than the fragment **6** of the previous case. When secured to the stem part **12**, the head part **14** preferably covers the portion **22** essentially or totally.

(75) In this case, once the securing positions are chosen, the sub-step of designing the stem part **12** is performed. The head part **14** may also be designed patient-specifically. In this designing sub-step, the stem part **12**, and optionally at least a part of the head part **14**, is designed so that the stem part **12** allows securing the fragments **4** and **5** and the head part **14** at the chosen securing positions. In other words, the designed stem part **12** enables or even imposes that, when the fragments **4** and **5** the head part **14** are secured thereto, they are in the planned positions.

(76) Then, during step C), the prosthesis **11**, including in particular the stem part **12** with the features designed in step B) above, is manufactured. If all or part of the head part **14** is designed patient-specifically during step B), the patient specific features of the head part **14**, or all the head part **14** is also manufactured during step C).

(77) The head part **14** may comprise a standard cap **50**. "Standard" means that the cap **50** is not patient-specific, although a cap of appropriate size and shape may be chosen among a definite collection of standard caps. Thus, the standard cap **50** is provided or manufactured separately from the patient-specific features manufactured at step C).

(78) The standard cap **50** comprises an articular surface **52** of convex shape, preferably spherical, for forming a prosthetic joint of the patient, combined with the glenoid cavity of the scapula of the patient, or any other auxiliary bone considered, in replacement for the surface **25** of fragment **6**. In other words, the cap **50** may be shaped as a dome as depicted on FIG. **1**. By "prosthetic joint", it is meant that at least a part of the joint is prosthetic. In this case, the standard cap **50** is a prosthetic part of the joint.

(79) Alternatively to a convex shape, depending on the situation, a concave shape may be used.

(80) The standard cap **50** also comprises a trunnion **55**, shown in dashed lines in FIG. **1**, protruding in an opposed to the surface **52**.

(81) The standard cap **50** also comprises a securing surface **54**, opposed to the surface **52** and formed at the end of the trunnion **55**. The standard cap **50** is configured to be secured to the stem

part **12** by means of the securing surface **54**.

(82) The head part **14** also comprises a patient-specific insert **56**. This insert **56** is patient specifically designed during step B) according to the data provided at step A), and manufactured during step C) according to this design. The insert **56** is a separate piece from the cap **50** and is assembled with it before or during surgery.

(83) The insert **56** is interposed between the cap **50** and the stem part **12**. More precisely, the insert **56** has a surface **57** bearing against a border surface **53** of the cap **50**. The border surface **53** surrounds the trunnion **55**, and is opposed to the surface **52**. The insert **56** also has an opposed surface **58**, that is configured to bear against the fracture lines **41** and **42** of the fragments **4** and **5**. The insert **56** is preferably shaped as a ring or a washer, as depicted on FIG. **1**, so that the cap **50** may be secured to the stem part **12** through the insert **56**. In this case, the trunnion **55** passes through a central hole of the insert **56**. The insert **56** is preferably fitted, for example conically fitted, onto the trunnion **55**.

(84) Alternatively, the cap of the head part **14** is made patient-specifically, designed during step B) and manufactured during step C). In this case, the cap and the insert may form a single piece instead of two distinct assembled pieces.

(85) During step B), the respective securing positions of the fragments **4** and **5** and of the head part **14** are chosen so that a respective chosen mechanical stress is applied to each fragment **4** and **5** by the head part **14**. In a preferable embodiment, the fragments **4** and **5** are compressed between the head part **14** and the fragment **2**. Alternatively, the head part **14** may apply mechanical stress on the fragments **4** and **5** without relying on the fragment **2**.

(86) More specifically, step C) may include patient-specifically designing the insert **56** so that the insert **56** applies the respective stress onto the fragments **4** and **5** with a chosen magnitude. In particular, the shape of the surface **58** may be designed in correspondence with the shape of the fracture lines **41** and **42**, so that the surface **58** may distribute the applied stress along the fracture lines **41** and **42**.

(87) Thus, when the prosthesis **11** is manufactured and implanted in the patient, a mechanical stress defined in advance patient specifically is applied to the reattached fragments **4** and **5**, and optionally to the fragment **2**, by the head part **14**, and optionally by the stem part **12**.

(88) The patient specific parts of the head part **14** may be obtained, during step C), by additive manufacturing. Preferably, the insert **56** is manufactured by additive manufacturing. If the cap **50** is patient-specific, the insert **56** and the cap **50** may both be obtained by additive manufacturing.

(89) Instead of having a patient specific surface **58**, the head part **14** may be chosen standard. Thus, only the relative securing position of the fragments **4** and **5** and of the head part **14** may allow defining the mechanical stress to be applied.

(90) After the prosthesis **11** is manufactured at step C) for the specific patient to be treated, the surgeon may proceed as follows for installing the prosthesis **11** in the patient's body.

(91) The surgeon firstly secures the stem part **12** onto the fragment **2**, by introduction of the coverable portion **16** into the cavity **8**. The surgeon checks that the mark **18** is superposed with the fracture line **9** and may adjust the position of the stem part if necessary. In the adequate position, the stem part **12** applies the chosen mechanical stress to the zone **40** of the cavity **8**.

(92) Secondly, the surgeon secures the fragments **4** and **5** onto the stem part **12**. For this purpose, the surgeon may have prepared the fragments **4** and **5** in advance, including drilling the blind holes. The blind holes may be drilled with help of a tool and/or a jig made patient-specifically during step C), designed at step B) based on the data of step A). In this case, the fragments **4** and **5** are positioned onto the stem part **12** with the plugs **30** and **31** inserted into the blind holes. The surgeon checks whether the positioning of the fragments **4** and **5** is correct by checking if the fragments **4** and **5** match with the mark **45**. If necessary, the position of the fragments **4** and **5** is adjusted by the surgeon with help of the mark **45**.

(93) Thirdly, the fragment **6** is secured to the stem part **12**. For this purpose, the surgeon may have

prepared the fragment **6** in advance, including drilling the blind hole. The blind hole may be drilled with help of a tool and/or a jig made patient-specifically during step C), designed at step B) based on the data of step A). The surgeon checks whether the positioning of the fragment **6** is correct by checking if the fragment **6** matches with the mark **45**. If necessary, the position of the fragment **6** is adjusted by the surgeon with help of the mark **45**. Securing the fragment **6** applies mechanical stress onto the fragments **4** and **5**.

(94) It can be provided that the fragments **4** and **5** are interposed between the fracture line **9** of fragment **2** and the fragment **6**. In this case, mechanical stress is also applied to fragment **2** by fragments **4** and **5**. In another embodiments, the fragments **4** and **5** are supported by the plugs **30** and **31**, onto which the mechanical stress is transmitted, without transmitting the mechanical stress to the fragment **2**.

(95) Fasteners, sutures, adhesive means or any suitable means may be used by the surgeon for maintaining the fragments **2**, **4**, **5** and **6** onto the stem part **12**.

(96) If the fragment **6** is to be replaced, said fragment **6** is not secured to the stem part **12**. Instead, the head part **14** is secured to the stem part **12**. Securing the head part **14** applies mechanical stress onto the fragments **4** and **5**, and optionally to the fragment **2**, in a similar manner.

(97) Fasteners, sutures, adhesive means or any suitable means may be used by the surgeon for maintaining the fragments **2**, **4** and **5** and the head part onto the stem part **12**.

(98) The external surface of at least one of the coverable portions **16**, **20**, **21** and **22** is preferably rough, or even comprises barbs, for helping securing the bone fragments **2**, **4**, **5** and **6** secured thereto.

Claims

1. A method of manufacturing a prosthesis for a fractured long bone of a patient, the method comprising the steps of: A) providing data representative of the fractured long bone of the patient, the fractured long bone comprising epiphyseal fragments, each epiphyseal fragment being: either a tuberosity fragment, secured to a muscle of the patient, the muscle being attached to the tuberosity fragment by means of a tendon of the muscle, or an articular fragment, being part of a joint of the patient, for articulating the long bone with an auxiliary bone of the patient; B) based on the data provided at step A), designing the prosthesis specifically to the patient, the prosthesis comprising a stem part configured for securing the epiphyseal fragments to the stem part, step B) comprising: a sub-step of choosing, specifically to the patient, respective securing positions of the epiphyseal fragments relative to each other, representative of how the epiphyseal fragments are planned to be positioned relative to each other when secured to the stem part, the securing positions being chosen so that a respective patient-specific mechanical stress is applied onto each epiphyseal fragment by at least one of the other epiphyseal fragments by securing the epiphyseal fragments to the respective chosen securing positions of the stem part, and a sub-step of designing the stem part so that the stem part is configured for securing the epiphyseal fragments at the respective chosen securing positions; and C) manufacturing the prosthesis including the stem part designed at step B).
2. The method according to claim 1, wherein step B) comprises the further sub-step of designing the stem part so that the stem part comprises a visible mark indicating the respective chosen securing positions of the epiphyseal fragments on the stem part.
3. The method according to claim 2, wherein step B) comprises the further sub-step of designing the stem part based on the data provided at step A), so that the stem part comprises, specifically to the patient, coverable epiphyseal portions, being configured for being covered respectively by the epiphyseal fragments when the epiphyseal fragments are secured to the stem part at the respective chosen securing positions; wherein the visible mark is designed so as to visually delimit the coverable epiphyseal portions from each other.
4. The method according to claim 1, wherein the stem part comprises, for at least one of the

epiphyseal fragments, a respective plug, for securing the concerned epiphyseal fragment to the stem part, said at least one plug applying mechanical stress onto the epiphyseal fragment secured thereto.

5. The method according to claim 1, wherein step A) includes using CT scans of the patient and step C) includes additive manufacturing of at least a part of the prosthesis.

6. The method according to claim 1, wherein the fractured long bone is a fractured humerus, the prosthesis being a shoulder prosthesis.

7. A method of manufacturing a prosthesis for a fractured long bone of a patient, the method comprising the steps of: A) providing data representative of the fractured long bone of the patient, the fractured long bone comprising: at least one viable tuberosity fragment, secured to a muscle of the patient, the muscle being attached to said at least one viable tuberosity fragment by means of a tendon of the muscle; and a damaged articular fragment, initially being part of a damaged joint of the patient, for articulating the long bone with an auxiliary bone of the patient; B) based on the data provided at step A), designing the prosthesis specifically to the patient, the prosthesis comprising a stem part, configured for securing the at least one viable tuberosity fragment to the stem part, and a head part, being configured to be secured to the stem part in replacement for the damaged articular fragment, step B) comprising: a sub-step of choosing, specifically to the patient, respective securing positions of the at least one viable tuberosity fragment and of the head part relative to each other, representative of how the at least one viable tuberosity fragment and the head part are planned to be positioned relative to each other when secured to the stem part, the securing positions being chosen so that a respective patient-specific mechanical stress is applied to each of the at least one viable tuberosity fragment by the head part by securing the at least one viable tuberosity fragment and the head part at the respective chosen securing positions of the stem part, and a sub-step of designing the stem part so that the at least one viable tuberosity fragment and the head part may be secured to the stem part at the respective chosen securing positions; and C) manufacturing the prosthesis including the stem part designed at step B) and providing or manufacturing the head part of the prosthesis.

8. The method according to claim 7, wherein the head part comprises: a standard cap, comprising an articular surface of concave or convex shape for forming a prosthetic joint for replacement of the damaged joint of the patient, and comprising a securing surface opposed to the articular surface, the standard cap being secured to the stem part by means of the securing surface; a patient-specific insert, designed during step B) patient specifically, and manufactured during step C), configured to be interposed between the standard cap and the stem part, and apply the respective mechanical stress onto each of said at least one viable tuberosity fragment.

9. The method according to claim 7, wherein step A) includes using CT scans of the patient and step C) includes additive manufacturing of at least a part of the prosthesis.

10. The method according to claim 7, wherein the fractured long bone is a fractured humerus, the prosthesis being a shoulder prosthesis.
